View results

Respondent

41 Anonymous

40:57

Time to complete

A. General information about the respondant

1. * Country

Mauritius

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

The Ministry of Tertiary Education, Science and Research

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://tertiary.govmu.org/SitePages/Index.aspx

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

The functions of the Tertiary Education and Scientific Research Division are:
· To transform Mauritius into a Knowledge Hub and a Centre of Excellence for Higher Learning which will serve the region and beyond.
· To translate the vision of the Ministry into implementation strategies in Tertiary Education, Science, Research and Technology.
· To design, review, implement and monitor educational policies, strategies and reforms in line with Government Programme.
· To prepare plans for the development of tertiary education sector with focus on access, quality, relevance, equity and achievement of all learners.
· To ensure that the Ministry is kept up to date with the latest educational trends in higher education and research sectors.
· To ensure optimum utilization of resources allocated to the higher education sector.
· To create an enabling environment for a higher education system that both generates and equips learners with innovative, cutting-edge knowledge and deep skills for increased competence in a dynamic work environment.
· Formulation of policies relating to Science and Technology
· Promotion of areas of Science and Technology with special emphasis on emerging areas
· Development of strategies to increase the enrolment in STEM/STEAM subjects for both genders.
· Organisation of research studies in Science and Technology
· Promotion of strategies to boost research towards innovation using Science and Technology

7. * Full name of officially designated contact person(s) who completed this survey

Prof (Dr) Kiran Bhujun

8. * Position of officially designated contact person(s) who completed this survey

Director (Tertiary Education and Scientific Research Division)

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Mr Navindsing Jugmohunsing

11. Brief description of the consultation process established for the preparation of the report

The consultation process for the preparation of the National Report on the implementation of the UNESCO Open Science recommendation went through a rigorous process aimed at fostering inclusive dialogue among stakeholders. Engaging a diverse array of participants, including researchers from both research institutions and higher education institutions, policymakers, and practitioners. The process facilitated the gathering of multifaceted perspectives that enrich the reports content.
By employing a combination of in presence face to face meeting and surveys. The process captured the voices from various disciplinary backgrounds to ensure a holistic understanding of open science dynamics in the Country.
Consequently, this methodology not only enhanced the credibility of the reporting but also serves to galvanize a collective commitment to further advancing open science principles in the Country.
Ultimately, the nuanced insights garnered from these consultations form the backbone of the report, enabling it to address key challenges and opportunities within the realm of open science, while also laying the groundwork for future research and policy initiatives.
At the preliminary stage, a National Committee comprising Ministry of Education, Tertiary Education, Science and Technology, Ministry of Communication, Technology and Innovation, Public Higher Education Institutions, Mauritius Research and Innovation Council and Mauritius Oceanography Institute was set up.
Different stakeholders presented their contribution in the implementation of the UNESCO Recommendation on Open Science and provided inputs as per the guidelines and questionnaire provided.
Ultimately, an iterative feedback mechanism was established during the consultation process to ensure that the contributions were not only heard but also integrated into the final report, enhancing its relevance and applicability in the field of Open Science
The National submitted its recommendation in the form of a draft Report to the Ministry of Education, Tertiary Education, Science and Technology who validated the Report.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

1. Ministry of Education, Tertiary Education, Science and Technology 2. The Ministry of Information Technology, Communication and Innovation 3. The University of Mauritius 4. The University of Technology, Mauritius 5. The Open University of Mauritius 6. The Université des Mascareignes 7. The Mahatma Gandhi Institute 8. The Mauritius Institute of Education. 9. The Mauritius Oceanography Institute 10. Statistics Mauritius 11. The Higher Education Commission 12. The Rajiv Gandhi Science Centre 13. Mauritius Research and Innovation Council

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

https://education.govmu.org/SitePages/Index.aspx, mitci.govmu.org/Pages/MITCI/Min.aspx, https://www.uom.ac.mu/, https://www.utm.ac.mu/, https://udm.ac.mu/fr/, https://www.open.ac.mu/, https://www.mgirti.ac.mu/, http://web.mie.ac.mu/, https://www.mric.mu/, https://moi.govmu.org/, https://rgsc.govmu.org/, statsmauritius.govmu.org/Pages/Statistics/statsbysubj.aspx, https://www.hec.mu/

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

none

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [x] Public
- [ ] Private
- [ ] Civil Society
- [ ] Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- [x] Yes
- [ ] No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The UNESCO Recommendation on Open Science has been shared with the following organisations: 1. University of Mauritius 2. University of Technology, Mauritius 3. Mahatma Gandhi Institute 4. Open University of Mauritius 5. Université des Mascareignes 6. Higher Education Commission 7. Mauritius Research and Innovation Council 8. Mauritius Institute of Education 9. Ministry of Tertiary Education, Science and Research 10. The Rajiv Gandhi Science Centre

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- [x] Yes
- [ ] No

19. 1.2 cont. **If yes,** which language(s)?

The languages used, spoken and written, in Mauritius are English (Official Language), French, Creole and several Asian Languages. Since the UNESCO Recommendation on Open Science is available in both French and English it is considered that it is accessible to the whole population.

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

    [i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- (•) Yes
- ( ) No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

At national level, the UNESCO recommendation on Open Science has been disseminated to relevant organisations who were requested to implement and report on status during stakeholders' meetings. Example of activities: 1. Capacity building for educators in science teaching 2. Community Engagement initiatives (refer to section 2.5) 3. National Science week (annual feature) 4. Seeking of inputs from Civil society and NGOs

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

- [x] Yes
- [ ] No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

The MRIC has set up a research repository to share all reports and findings, subject to IP rules, emanating from funding schemes; the reports are available to the public (https://www.repository.mu/mrc/). At this date, a total of 295 publications have been uploaded for the theme Natural Sciences, 215 publications have been uploaded for the theme Engineering and Technology, 199 publications for the theme Social Sciences, 61 publications for the theme Agricultural and Veterinary Sciences, 48 publications for the theme Medical and Health Sciences and 5 publications for the theme Humanities and the Arts (Source: Mauritius Research Repository; https://www.repository.mu/mrc/) Science and research outputs from the University of Mauritius are available to the public. Journals Staff FT Staff PT Student UoM Guest Public UoM Lib Manager Other TEI Admin Other TEIs Registered MPhil Pre Registered MPhil Pre Registered by research LPVC MinistryOrg 1 DOAJ □ □ □ □ □ □ □ □ □ □ □ □ □ 2 JSTOR □ □ □ □ □ □ □ □ □ □ □ □ □ 3 Emerald □ □ □ □ □ □ □ □ □ □ □ □ □ 4 UNESCO EOLSS □ □ □ □ □ □ □ □ □ □ □ □ □ 5 UNCTAD □ □ □ □ □ □ □ □ □ □ □ □ □ 6 UpToDate Medical e-Resource □ □ □ □ □ □ □ □ □ □ □ □ □ 7 ScienceDirect □ □ □ □ □ □ □ □ □ □ □ □ □ 8 HSTalks (The Business & Management Collection) □ □ □ □ □ □ □ □ □ □ □ □ □ Resources Staff FT Staff PT * Student UoM Guest Public UoM Lib Manager Other TEI Admin Other TEIs Registered MPhil Pre Registered MPhil Pre Registered by research LPVC MinistryOrg 1 Books □ □ □ □ □ □ □ □ □ □ □ □ □ 2 eBooks □ □ □ □ □ □ □ □ □ □ □ □ □ 3 eJournals □ □ □ □ □* □ □ □ □ □ □ □ □ 4 Dissertation □ □ □ □ □ □ □ □ □ □ □ □ □ 5 Past Exam Paper □ □ □ □ □ □ □ □ □ □ □ □ □- access denied □ – access granted □* – limited access granted - * Part-time staff- (Access granted to online resources provided they have UoM credentials). UdM: Starting in February 2025, the university will launch a series of research seminars held every three months, specifically designed to champion the principles of open science. These seminars will provide an engaging platform for researchers, academics, and students to explore and embrace the values of transparency, collaboration, and inclusivity in their scholarly work. Each session will feature expert speakers, practical workshops, and interactive discussions aimed at fostering a deeper understanding of open science methodologies, including open data sharing, reproducibility, and open access publishing. MIE Research Journal of Education available on the MIE website [https://web.mie.ac.mu/research-consultancy/] Science curriculum textbooks for Primary and Lower Secondary available on the MIE website [https://web.mie.ac.mu/primary-curriculum/] [https://web.mie.ac.mu/secondary-regularprogramme/] OUM: (i) The development of an OER manual for the module Business for Sustainable Development. This OER is used by all MBA learners who are registered for the CEMBA and CEMPA COL programme worldwide. (ii) 12 academics from the OUM were nominated to create an OER course delivered to our learners. 12 OERs were produced and provided to our learners. This was done with the support of HEC and COL. The modules are: ü Teaching and Learning in the Digital Age ü Company Law and Corporate Reconstruction ü Operational Research ü Life Skills for educators ü Studying Literature ü Information system and Security ü Quality Management ü Marketing Planning ü Understanding Prose ü Organisational Behaviour and analysis ü Marketing Management ü Risk Management. HEC - A total of 88 ODL/OER courses were developed under the TEL Project to be uploaded on the OER repository (Please refer to Sections 2.5 and 2.6 for more details). University of Technology, Mauritius (UTM) A repository on research projects, publications and research activities is available at the University of Technology (UTM). The UTM is envisaging the publication of a yearly research bulletin. All Academic Staff have a Google Scholar Profil

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

- (selected) Yes
- No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Mauritius is currently working on a strategy on S&T, efforts are underway to develop the STI sector. The country is investing in a collaborative research and development grant programme to encourage innovative ideas, commercialise them, create jobs, and ultimately generate wealth. https://doi.org/10.1007/978-981-19-6802-0_11 The National Roadmap of MRIC contains insights into STI and with orientation toward industries Roadmap link: https://www.mric.mu/_files/ugd/f94712_c1b1a364bd284fa8ac9273fe423211a3.pdf

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- [x] open access to scientific knowledge
- [ ] open science infrastructure
- [ ] open engagement of societal actors
- [x] open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ○ Yes
- ○ Partly
- ◉ No

28. 2.2 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

No Framework on open science exists. However, there is desire to develop such a policy across the HEIs but the difficulty lies on who will coordinate such an initiative so that there is consensus. The applicability and implications of IP rights for Open Science practices involving various HEIs in Mauritius should priorly be discussed among stakeholders and policy makers.

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ○ Yes
- ○ Under development
- ◉ No

30. 2.3 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

The development of such a policy document on open science is ongoing.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

- ○ Yes
- ○ Under development
- ◉ No

32. 2.4 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

No No framework on open science exists. However, there is desire to develop such a policy. [Currently, there is no established framework on open science. However, there is a clear and growing desire to create such a policy. This enthusiasm represents an exciting opportunity to shape a forward-thinking approach that encourages transparency, collaboration, and accessibility in research and innovation. By developing a robust open science framework, stakeholders can promote equitable knowledge sharing, foster interdisciplinary collaboration, and empower researchers and HEIs to address global challenges more effectively.]

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

- (•) Yes
- ( ) Under development
- ( ) No

34. 2.5 cont. **If yes,** please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9

The national OER policy for Mauritius is titled, Mauritius National Open Educational Resources Policy. The policy aims to provide direction in using OER to support quality teaching and learning in the Mauritian education and training system. A High-Level Committee for Monitoring the Mauritius National Open Educational Resources Policy has been set up at the level of Higher Education Commission. The Higher Education Commission provides funding to support researchers in their projects, and the findings are available to all researchers. While there are no existing policies, many of our research projects indirectly encourage public participation. For example, several initiatives and strategies promote open science in our country: 1. Research Knowledge Dissemination: • Regular half-day workshops at the Mauritius Research and Innovation Council (MRIC) for sharing research outcomes • Organization of stakeholder and validation workshops to ensure research transparency and community engagement • National Innovation Challenge programs (BR SH) to promote innovative research solutions 2. Open Access and Research Communication: • Support for Open Access publications including funding for publication fees • Regular research dissemination workshops and forums like "Assises de L'Agriculture" 3. Environmental Research and Community Engagement: • Coral Restoration Workshop involving University of Mauritius students • Butterfly Garden Project on the University Farm • Climate Smart Agriculture training programs for farmers • Year-long Food Waste Sensitization campaign • Food waste reduction initiatives 4. Inclusive Research Participation: • Special programs supporting Women Entrepreneurs in research and innovation • Initiatives supporting Rodriguan students in research activities Input from RGSC The Rajiv Gandhi Science Centre organises various competitions for secondary school students, including: • Model Glider- students of Grade 10 and 12 construct and fly their gliders • Science Quest- students from grade 7 to 13 prepare and implement a project proposal on real-world problem • Debat Klima- the contests target secondary students from grade 10 to 12 debate on climate change topics in Creole. These events aim to: • Promote STEM education • Foster creativity, critical thinking, and scientific curiosity • Encourage young minds to engage with science and technology • The competitions provide an interactive and competitive environment for students to explore and learn. RGCS organises various events including the following: • Seminars: International Day for Biodiversity- to raise awareness about biodiversity conservation. • Workshop on Robotics: Introducing students to robotics and fostering interest in STEM. Outreach Activities: □ Career Guidance Talks: Inspiring future scientists and guiding students on STEM careers. □ Sky Observation: Spark curiosity about astronomy through hands-on sessions. □ National Science Week: Organising various events at the national level to celebrate science. □ Talks and Science Activities at Community Centres: Engaging the local community in science education. □ Science Shows in Primary and Secondary Schools: Promoting science through engaging demonstrations. □ Robotic Demonstrations in Schools: Showcasing robotics to inspire students and encourage learning. □ Workshop for educators on hands-on approach of teaching □ Open days at the Centre UdM: The module RESPONSABLE.S is designed to cultivate a culture of responsible citizenship and ethical practices in research and academia. In the context of open science, the module emphasizes the principles of transparency, inclusivity, and collaboration in the creation and dissemination of knowledge. It introduces learners to concepts such as data sharing and research reproducibility, aligning with global trends toward the democratization of knowledge. By equipping students with the necessary skills and understanding to engage in open science practices, the

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- (selected) Yes
- Under development
- No

36. 2.6 cont. **If yes or under development,** please provide details and links as relevant.

Mauritius has implemented specific funding mechanisms to support open science through the Open Educational Resources (OER) Policy. This policy provides support for publicly funded institutions to align their resources and initiatives with UN Sustainable Development Goal 4 and related SDGs. Additionally, private higher education institutions are encouraged to leverage this framework by adopting the national OER Policy, thus further enabling the creation and dissemination of OER materials. Publicly funded educational institutions and autonomous bodies are guided by this policy to channel resources effectively toward open educational initiatives. The Higher Education Commission provides funding for researchers at higher education institutions (HEIs) to undertake research projects. Additionally, public research HEIs also offer financial support to their faculty members who are involved in research.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

- (selected) Yes
- No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

Some publications emanated from HEC's funded projects are published in Open Access journals and with the development of an open science policy, this element will be captured. MRIC publish all reports of funded projects on a Research Repository open to the public. There is also wide dissemination of funded projects to the public at large The University of Mauritius has policies that require researchers to consider open science practices when applying for internal research funding. This includes commitments to open access publishing and data sharing. In addition, open science practices have been integrated into existing research funding mechanisms through several channels: • Higher Education Commission (HEC) website provides transparent access to research funding opportunities and guidelines • University of Mauritius (UoM) website features research funding mechanisms and open science requirements • Regular workshops organized to inform researchers about funding opportunities and open science requirements • Research Week (RW) organization with internal funding support to promote research dissemination • Internal funding mechanisms that incorporate open science principles • Dedicated resources for research dissemination and knowledge sharing. These mechanisms demonstrate our commitment to integrating open science practices into the research funding landscape, ensuring transparency and accessibility of research outputs. Sharing of Open Scientific Knowledge is done e.g. Research Symposiums, Conferences (both local and international) and the UoM Research Week is held yearly. UoM works in close collaboration with industry, NGOs, Ministries, International Organisations by way of Research Collaborative Agreements and Memoranda of Understanding

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

- (selected) Yes
- No

40. 2.8 cont. **If yes**, please provide details and links as relevant.

• The Higher Education Commission is compiling a List of Potentially Impactful Research (LPIR) to promote collaboration between public higher education institutions and NGOs. • Funding of joint research projects involving public and private partnerships. There are some efforts that have been made for public-private partnerships on open science but such partnerships are not structured and not documented, but on an adhoc basis. The MRIC has Research and Innovation funding schemes promoting equal fund contributions from public and private sectors. Efforts are underway to foster more equitable public-private partnerships on open science. The Mauritius Research and Innovation Council (MRIC) has initiated programs that encourage collaboration between public research institutions and private companies. These programs aim to leverage the strengths of both sectors to advance open science practices. • MRIC has a funding mechanism designed to stimulate collaboration between public research institutions and private companies. This initiative aims to ensure that research outputs are accessible and that both sectors can work together effectively while prioritizing public interest. The establishment of Public Research Entities (PREs) where academia, industry, and government collaborate on research projects. This collaborative approach ensures that research outputs are accessible and that both sectors can work together effectively while prioritizing the public interest In addition, efforts have been made and are planned to foster equitable public-private partnerships on open science, including: 1. University-led Initiatives: • The 2nd Edition of University of Mauritius Research and Innovation Week, promoting collaboration between academic and private sector stakeholders • Special events planned for the UoM's 60th Anniversary celebrations, creating platforms for public-private engagement in research • Research Weeks, which provide opportunities for: Public-private sector networking Research collaboration development of Knowledge sharing between academic and industry partners. • Innovation Week: Bringing together researchers, private sector partners, and public stakeholders to showcase research and foster collaborations • Doctoral Colloquium: Providing opportunities for: PhD researchers to present their work to potential industry partners of Networking between academic and private sector stakeholders Development of collaborative research projects Inputs from UdM and UTM UdM: There is a strong commitment, starting in February 2025, to initiating regular research seminars on campus specifically designed also to promote and advance the principles of open science. These seminars will serve as vibrant platforms for fostering a culture of transparency, collaboration, and accessibility in research practices. They aim to equip researchers, students, and faculty members with the knowledge and skills to embrace open data sharing, reproducibility, and interdisciplinary collaboration. By encouraging active engagement with open science methodologies, these seminars will drive innovation, enhance the visibility of research outputs, and position the institution as a leader in the global movement towards democratising knowledge. OUM The Open University does collaborative research with other local institutions. The output of the studies is disseminated to the general public but not open access. University of Technology, Mauritius (UTM) The University has collaboration for joint programmes where students can conduct research using open access sources. Such partnerships are as follows: · Joint and dual programmes with Monash University · Student presentation to The Raj H. Prayag Award and Gold Medal initiative · Partnerships with Scotts & Co Ltd for Apprenticeship and CPD courses in pharmacy · Guest talk from MoU with Objectivity Software Services Ltd and of joint workshop on Software development, Agile Methodology, and Cyber Security & Testing

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ○ Yes
- ◉ Under development
- ○ No

42. 2.9 cont. **If yes or under development,** please provide details and links as relevant.

Ongoing

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

0.23% (Figure has been calculated by UNESCO Institute for Statistics (UIS) and sent to the MRIC)

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Internet access and use by household and individuals, 2020-2023 20201 20212 2022 20232 Percentage of households with internet access 72.6 73.8 75.03 76.5 Percentage of individuals (aged 5 years and above) using the internet 67.7 71.6 75.52 79.5 1 source: Continuous Multi-Purpose Household Survey (CMPHS 2020) 2 Estimates submitted in ITU Questionnaires 3 Source: Housing & Population Census 2022

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- ○ Yes
- ◉ No

46. 3.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

From a simple check on Wikipedia, we can count over 140 NRENs worldwide 2024. Sadly, a Mauritian NREN is not (yet) on the list. Discussions however to initiate such a network started around October 2015 with a small team of academics. Presentations and documents were shared with the Ministry of Education since then, culminating in the visit of the CEO of the UbuntuNet Alliance in December 2019. During his visit, along with a strong delegation, a number of initiatives were taken including a full-day workshop on the importance of NRENs (where the Minister of Education was present) and a high-level round-table committee meeting hosted at the Ministry of Education with the attendance of representatives of the Ministry of TCI and Ministry of Finance. The meeting ended on a very positive note with all stakeholders agreeing to the importance of an NREN for the country. The UbuntuNet Alliance has obtained funding of the order of 16 million USD to enable the set-up of new NRENs in Africa. Mauritius forms part of a list of contenders for this funding which will cover about 80% of capital costs (totaling about 2 million USD) for the NREN connection to the UbuntuNet Alliance network. The other 20% (or some 400,000 USD) should be provisioned by the country and could be sourced from Government funding or other means. The main obstacles for the setting of the Mauritian NREN are: 1. Financial – the 20% from Government funding as well as the financial model for the NREN for its sustainability 2. The Technical aspects in terms of the type of technology to be sued for the inland connectivity. There are no dedicated NRENS however, there are a repository of research reports available on the MRIC website, accessible to the public. However, not all reports are fully available. Regarding digital archiving and public access to research, the University of Mauritius (UoM) has initiated efforts in this area. The UoM Library also provides access to various digital resources and repositories. Access matrix has been inserted in section 1.3

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ○ Yes
- ○ Under development
- ◉ No

48. 3.4 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

MRIC Research Repository: Mauritius Research Repository Not yet Currently, there are no established open science infrastructures. The main reasons may be attributed to the following: • It could be that there has been insufficient financial investment directed toward building and maintaining open science facilities, which limits our ability to create robust systems that support research collaboration and data sharing. • Existing regulations often hinder effective collaboration between public and private sectors, making it difficult to establish comprehensive infrastructures that facilitate open science practices. • Current research infrastructures tend to be fragmented, lacking coordination and integration necessary for effective open science practices. However, at the University of Mauritius, there are some-level of digital infrastructures in place: • Research Management Information System at the University of Mauritius for managing research data and outputs • Project-specific websites (e.g., DESIRA website) providing access to research information and results • Subscription to Elsevier, providing researchers access to: o Scientific publications o Research databases of Academic resources

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- (selected) Yes
- No

50. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

Currently, we do not host or fund any federated regional or international information technology infrastructure for open science. While there are no national infrastructures, some existing regional and international open science infrastructures are accessible to us. E.g CIRAD, FAO databases.

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- (selected) Yes
- No

52. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

Our Faculty is involved in some regional and international research networks through faculty participation in: • RUFORUM (Regional Universities Forum for Capacity Building in Agriculture) • FANRPAN (Food, Agriculture and Natural Resources Policy Analysis Network) • H3BIONET (Pan African Bioinformatics Network for H3Africa) • SANOI (Southern African Network of Oceanographic Institutions) • PreRAD (Partnerships for Research and Development) • REAP (Research and Education Network) These collaborations facilitate: • Shared research infrastructure • Knowledge exchange • Capacity building • Joint research initiatives • Access to regional research platforms • South-South and North-South research partnerships UdM: UdM’s collaboration with Université de Limoges, France, provides a valuable opportunity to align research initiatives with the principles of open science. This partnership facilitates the exchange of knowledge, expertise, and resources between institutions, creating a foundation for interdisciplinary collaboration and innovative research practices. Through this collaboration, researchers adopt open data-sharing practices, co-publish findings in open-access journals, and develop frameworks for equitable access to scientific knowledge. The partnership also fosters capacity-building by organising workshops, seminars, and training sessions focused on open science methodologies, empowering both institutions to lead by example in democratising research. MIE: Yes Joint research collaboration on STEM between MIE and University of KwaZulu Natal (South Africa) and between MIE and Dublin University. Collaboration between MIE and University of Bordeaux on Movement and Physical Education. Collaboration between MIE and University of Tours, France Joint research collaboration between MIE and Homi Bhabha Centre for Science Education, India MIE collaboration with UNESCO and OCED on Climate Change Education Research collaboration between MIE and University College of London (UK) Organisation of the Distance Education and Teachers' Training in Africa (DETA) Conference 2025 by MIE in collaboration with University of Pretoria.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

53. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ○ Yes
- ◉ No

54. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Inadequate infrastructure or strategic support, implementing open science practices and training programs can be challenging, slowing down or preventing systematic capacity-building efforts. Currently, there has been no systematic capacity building on open science concepts and practices within higher education and research-performing institutions.

55. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- ○ Yes
- ◉ No

56. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Ongoing

57. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

- ○ Yes
- ◉ No

58. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

> Currently, there is no explicit framework of open science competencies incorporated into higher education research skills curricula.

59. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ○ Yes
- ○ Partly
- ◉ No

60. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ○ Yes
- ◉ No

61. 4.5 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

62. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- ◉ Yes
- ○ No

63. 5.1 cont. **If yes**, please provide more information and links as relevant.

We currently assess the sustainability of our research projects, and are looking into integrating open science principles into our evaluation processes. HEC- At the level of HEC, we have reviewed the Research Publication Incentive Scheme to consider publications in open-access journals.

64. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- ○ Yes
- ● No

65. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

66. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

- ● Yes
- ○ No

67. 5.3 cont. **If yes,** for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

This will be taken care during the forthcoming National Research Week 2025 (end of April)

68. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

- ○ Yes
- ◉ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- ○ YES, at the national level
- ○ YES, at the institutional level
- ◉ YES, at both the national and institutional levels
- ○ No

70. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- [x] promotion of pre-prints
- [x] exploration of open peer-review practices
- [ ] valuing and sharing of negative scientific results and associated data
- [x] participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- [ ] development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- [ ] promoting open innovation practices
- [ ] Other

71. 6.1. cont. **If yes**, please provide more information and links as relevant

There have been initiatives at both national and institutional levels that promote innovative and participatory methodological changes throughout the research cycle to enhance the openness of the scientific process. Many academics publish in open access journals, facilitating wider dissemination of their research. Additionally, in some research projects, stakeholders are actively involved, contributing to the research process and ensuring that diverse perspectives are considered. These efforts reflect a commitment to fostering openness and collaboration in scientific research. There has also been booklets created for a wider public audience, aimed at making the research more accessible and engaging to non-specialists. At the level of the MRIC, we only have dissemination after completion of MRIC Funded projects. The outcomes of projects are regularly disseminated to stakeholders through publications and events. UdM: Collaboration in inter-institutional research projects funded by the Higher Education Commission and other local/international organisations, like HSBC, offers a significant opportunity to promote data sharing and advance open science principles. These collaborative efforts bring together diverse expertise, perspectives, and resources from multiple institutions, creating a dynamic environment for conducting high-impact research. UTM HEC- A Standard Operating Procedure between HEC and MRIC is being prepared and the above Section 6.1 will be discussed with the MRIC. RGSC Conducted National Youth Conference in year 2018 and 2019 and published a consolidated publications of the research articles. Publised book chapter on STEM Education at Preprimary level Conducted seminar on Science Eduaction for educators at secondary level and recommendations submitted to the Ministry of Education

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

72. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

At the level of HEC, we have a scheme titled 'Recruitment of International Faculty' to promote and enhance international scientific collaborations, using the concept of open resources. While many academics strive to engage in international research collaborations, these efforts often occur without fully embracing the necessary practices of open science. Nevertheless, the ongoing initiatives reflect a commitment to fostering international scientific collaborations that align with open science principles. By encouraging transparency, inclusivity, and cooperation among diverse stakeholders, we are working towards a more equitable and accessible scientific landscape that benefits both science and society as a whole. MIE- International research collaborations. See section 3.6

73. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- (selected) Yes
- No

74. 7.2 cont. **If yes**, please provide links as relevant and more information.

There is recognition of the importance of fostering collaboration in open science, and discussions are ongoing about the potential for developing a comprehensive strategy in the future. Efforts to raise awareness and build capacity in open science practices must be considered as steps towards enhancing international cooperation.

75. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

- Yes
- (selected) No

76. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

We are not aware of any discussions regarding the establishment of regional and international funding mechanisms for open science. This lack of awareness may be due to limited communication about ongoing initiatives or the absence of active participation in such discussions. However, we recognize the importance of funding mechanisms for promoting open science and hope to explore opportunities for involvement in future discussions. OUM: The IORAG newsletter on its activities regarding fisheries management, maritime safety and security, trade and investment facilities, disaster risk management, academic sciences and technology cooperation are available to the public on its website. The activities are also available on Tiktok and Youtube. It also organises webinars open to the public. MIE- Funding of collaborative research undertaken by MIE with other foreign institutions such as UCL and UKZN

Part 8. GENERAL CONSIDERATIONS

77. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- (selected) Yes
- No

78. 8.1 cont. **If yes**, please provide more information.

While Mauritius is making progress in integrating open science principles, developing a cohesive national strategy or agenda would further enhance its ability to fully embrace UNESCO's recommendations. Continued participation in international dialogues and collaborations will be crucial for advancing these efforts. However, there are challenges that may impede the full implementation of an open science agenda. • Infrastructure costs for digital platforms and data storage • Limited human resources for managing open science initiatives • High bureaucratic mechanisms • Limited awareness of open science practices • Traditional research culture resistant to open sharing • Lack of standardized protocols for data sharing • High costs associated with journal subscriptions • Need for more trained personnel in data management • Limited expertise in open science practices • Need for more technical support staff MIE Too fastidious bureaucratic procedures to follow to engage in open science with other countries. Financial constraints Lack of expertise.

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

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